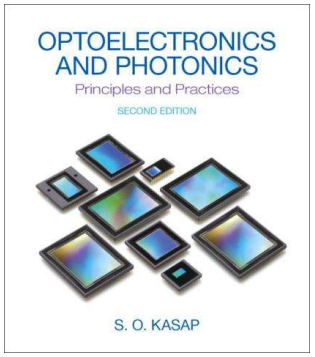
Chapter 3 Semiconductor Science and Light-Emitting Diodes



- Ref-2: Ch.1: Basic Semiconductor Properties

3.1 Review of Semiconductor Concepts and Energy Bands

- Ref-2: Ch.3-A: Energy Band Theory
- Ref-2: Ch.3-B: Energy Band Theory

3.4 Direct and Indirect Bandgap Semiconductors: E-k Diagrams

3.2 Semiconductor Statistics

3.3 Extrinsic Semiconductors

- Ref-2: Ch.4 : Equilibrium Carrier Statistics

3.5 pn Junction Principles---중간고사---

3.6 pn Junction Reverse Current

3.7 pn Junction Dynamic Resistance and Capacitances

3.9 pn Junction Band Diagram

3.8 Recombination Lifetime

Ref-2:Ch5: Recombination-Generation Processes-1/2

Shockley–Read–Hall (SRH) recombination

3.10 Heterojunctions

3.11 Light-Emitting Diodes: Principles

3.12 Quantum Well High Intensity LEDs

3.13 LED Materials and Structures

3.14 LED Efficiencies and Luminous Flux

3.15 Basic LED Characteristics

3.16 LEDs for Optical Fiber Communications

3.17 Phosphors and White LEDs

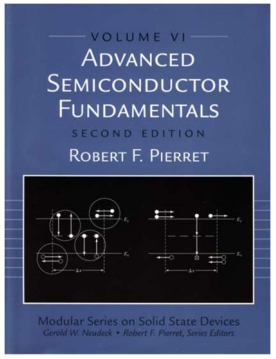
+ Advanced Semiconductor Fundamentals:
2nd Ed. Rober F. Peirret

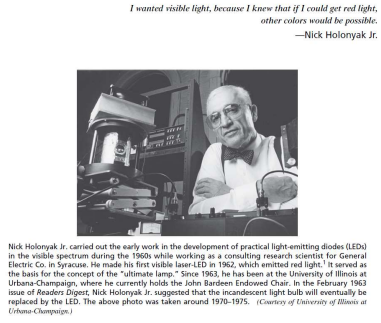
Ch.1: Basic Semiconductor Properties

Ch.3: Energy Band Theory

Ch.4: Equilibrium Carrier Statistics

Ch 5: Recombination-Generation Processes

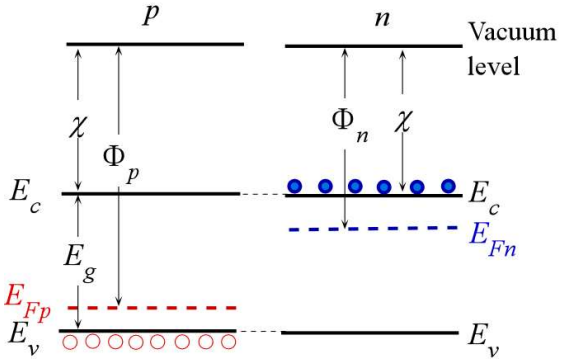




3.10 Heterojunctions

◆ Homojunction:

Junction between **the same crystals**
with **the same Bandgap**

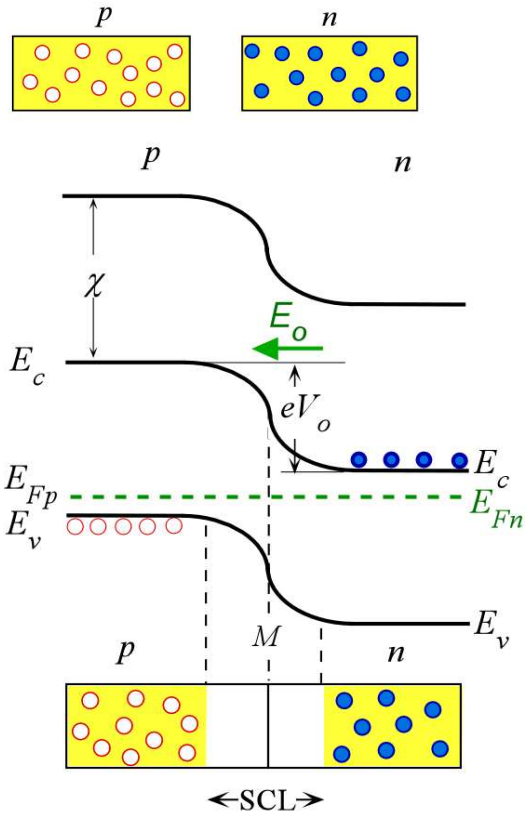


➢ Bandgap difference:

$\Delta E_g = 0$

$\Delta E_g = E_{g2} - E_{g1} = 0$

→ $\Delta E_c + \Delta E_v = 0$



➢ Band offsets:

$\Delta E_c = 0$, $\Delta E_v = 0$:

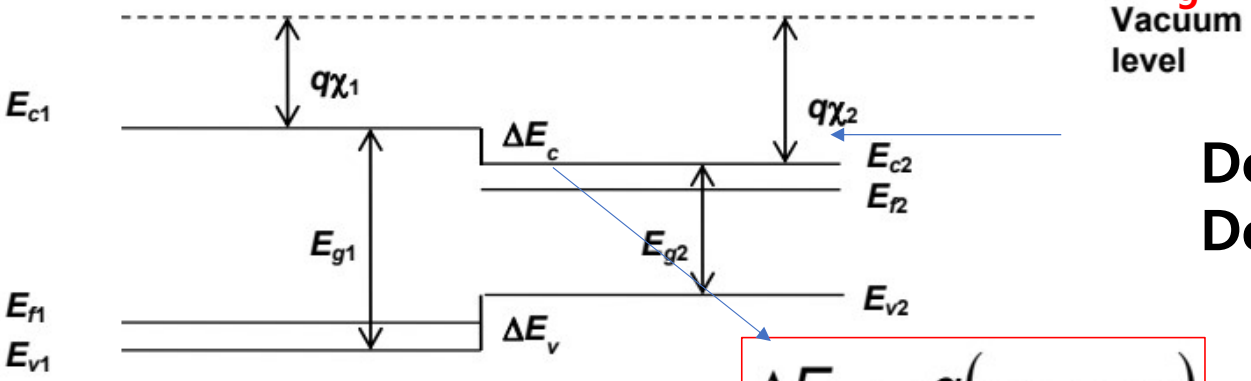
$\Delta E_c = E_{c2} - E_{c1} = 0$,

$\Delta E_v = E_{v2} - E_{v1} = 0$

3.10 Heterojunctions

◆ Heterojunction/Heterostructures:

Junction between two **different** semiconductor crystals
with **different bandgaps** (E_{g1}, E_{g2})



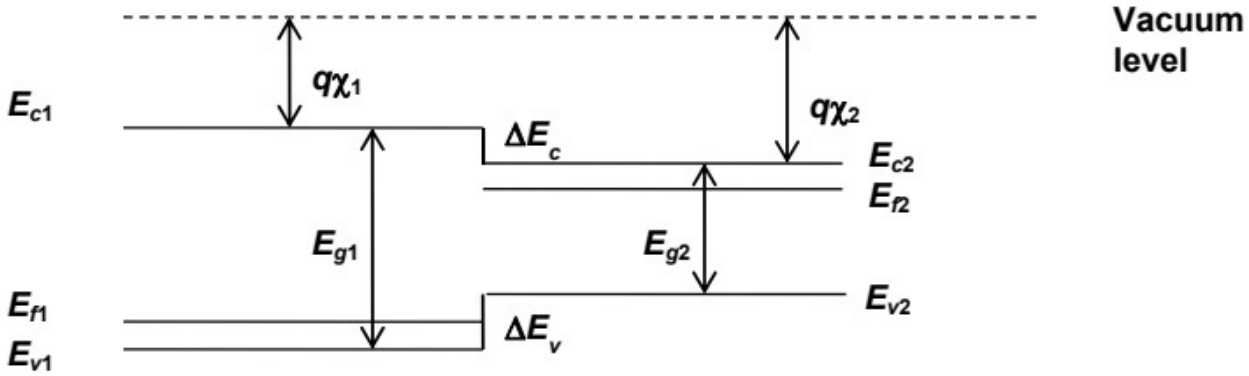
$$\Delta E_c = q(\chi_2 - \chi_1)$$

Doping in the wider bandgap: N, P
Doping in the lower bandgap: n, p

◆ Electron affinity rule (Anderson's Rule) for band alignment:

- At a heterojunction between different semiconductors the **relative alignment of bands** is dictated **by their electron affinities**
- **The conduction band offset at a heterojunction interface**
  = **The difference in the electron affinities between the two semiconductors**.

Electron Affinity Rule (Anderson's Rule) and Band Alignment:



- **Electron affinities difference = Conduction band offset :**

$$\Delta E_c = q(\chi_2 - \chi_1)$$

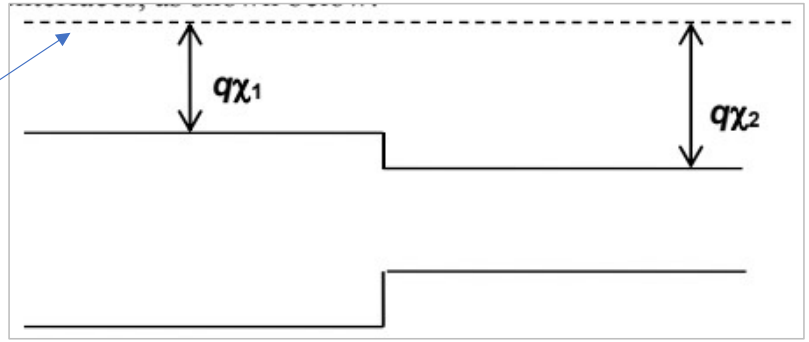
- **Bandgap difference ($\Delta E_g \neq 0$):**
  - ➔ $\Delta E_g = E_{g2} - E_{g1} = \Delta E_c + \Delta E_v$
- **Band offsets ($\Delta E_c \neq 0$, or $\Delta E_v \neq 0$):**
  - ➔ $\Delta E_c = E_{c2} - E_{c1}$
  - ➔ $\Delta E_v = E_{v2} - E_{v1}$

- **Electron affinity**
  - $Al_xGa_{1-x}As$: 4.07 – 1.1 x (eV) for $0 \leq x \leq 0.45$ (Direct bandgap region)
  - GaAs: 4.07 eV

◆ Three Type Heterojunctions:
Electron Affinity Rule and Band Alignment:

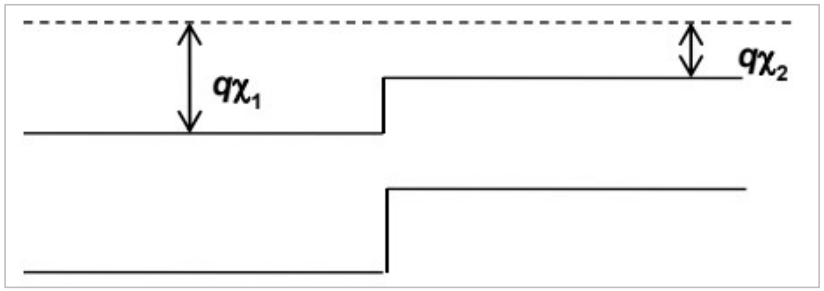
◆ **Type I** interface (**Straddled** gaps)

Vacuum level



Type I: Straddled bandgap alignment heterojunction
- $GaAs/Al_xGa_{1-x}As$,
- Ga_xIn_{1-x}/InP,
- $GaAs/Ga_xIn_{1-x}P$

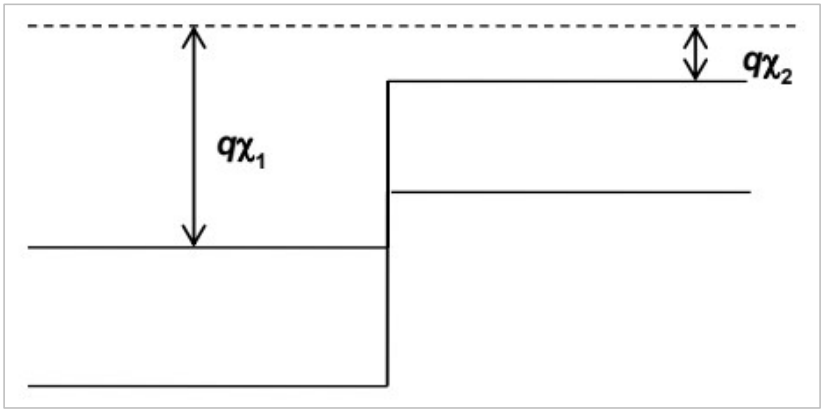
◆ **Type II interface (Staggered gaps)**



Type II: Staggered lineup heterojunction
- $Ga_xIn_{1-x}As/GaAs_ySb_{1-y}$

◆ **Type III interface (Broken gaps)**

❖ **The affinity rule does not always work well**



Type III: Broken gaps
- InAs/GaSb,
- HgTe/CdTe

Often the conduction band offsets ΔEc at interfaces are determined experimentally.

Heterojunctions

Band alignment and a P-n(Pn) heterojunction

Assume that:

(1) A **type-I** interface

(2) The **known conduction band offset** ΔE_c

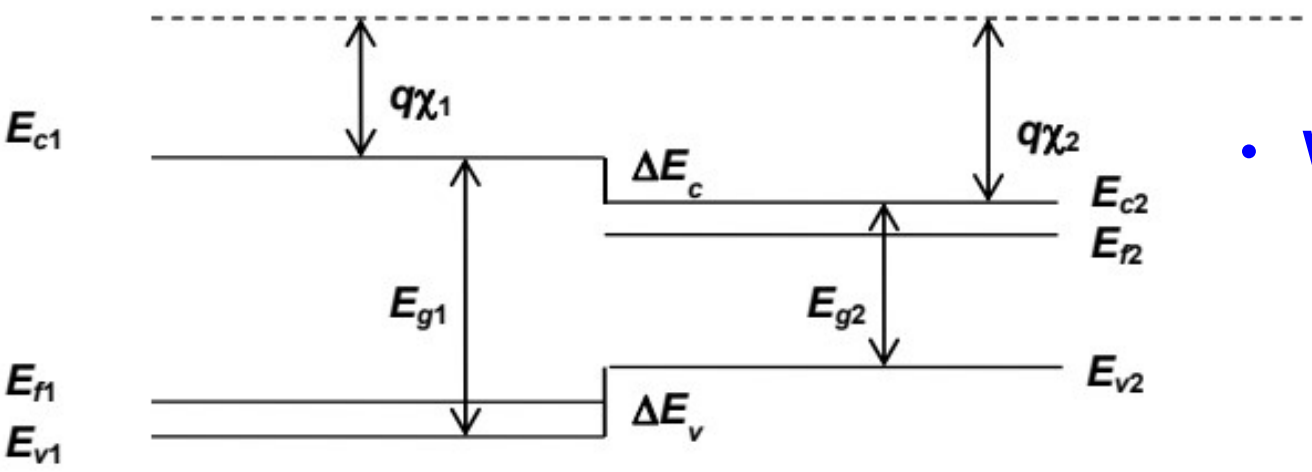
(3) The doping in semiconductor-1: **P-type** (N_a)

(4) The doping in semiconductor-2: **n-type** (N_d)

(5) All dopants: Ionized.

Doping in the wider bandgap: N, P

Doping in the lower bandgap: n, p

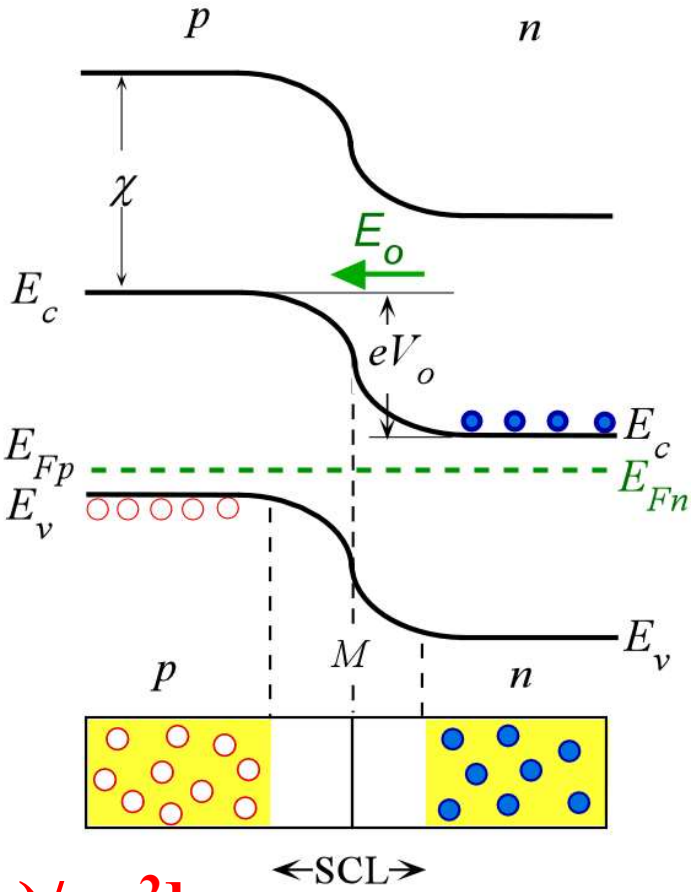


- **Work Functions: Φ_{p1} Φ_{n2}**

$$\Phi_{p1} = q\chi_1 + (E_{C1} - E_{F1})$$
$$= q\chi_1 + E_{g1} - (E_{F1} - E_{v1})$$

$$\Phi_{n2} = q\chi_2 + (E_{C2} - E_{F2})$$

A pn Homojunction Diode in Thermal Equilibrium:



- **On the *n*-side**

$$n = N_c \exp[-(E_c - E_{Fn})/k_B T] = N_d$$

$$\rightarrow E_c - E_{Fn} = -k_B T \ln(N_d / N_c) \quad (1)$$

- **On the *p*-side**

$$n = N_c \exp[-(E_c - E_{Fp})/k_B T] = n_i^2/N_a$$

$$\rightarrow E_c - E_{Fp} = -k_B T \ln[n_i^2 / (N_a N_c)] \quad (2)$$

➢ **Built-in Voltage: Subtracting Eq. (2) from Eq. (1)**

$$\rightarrow eV_o = E_{Fn} - E_{Fp} = k_B T \ln[(N_a N_d)/ n_i^2]$$

A Pn Heterojunction Diode in Thermal Equilibrium:

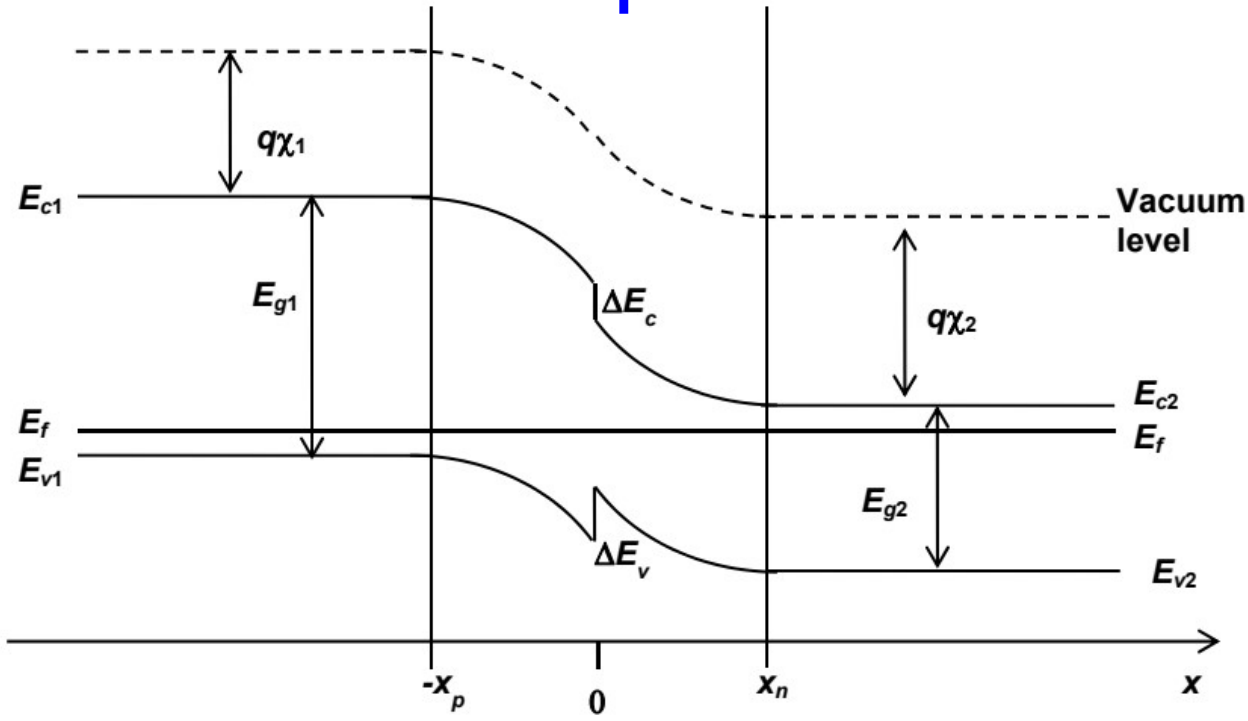
➢ Built-in potential V_{bi}:

$$qV_{bi} = |E_{F1} - E_{F2}|$$

$$qV_{bi} = \Phi_{p1} - \Phi_{n2}$$

$$\Phi_{p1} = q\chi_1 + (E_{C1} - E_{F1})$$
$$= q\chi_1 + E_{g1} - (E_{F1} - E_{V1})$$

$$\Phi_{n2} = q\chi_2 + (E_{C2} - E_{F2})$$



$$qV_{bi} = (q\chi_1 - q\chi_2) + E_{g1} - (E_{F1} - E_{V1}) - (E_{C2} - E_{F2}) \qquad E_{C2} - E_{F2} = -k_B T \ln (N_{D1} / N_{C2})$$

$$V_{bi} = (\chi_1 - \chi_2) + \frac{E_{g1}}{q} - \frac{kT}{q} \ln\left(\frac{N_{V1}}{N_{A1}}\right) - \frac{kT}{q} \ln\left(\frac{N_{C2}}{N_{D2}}\right) \qquad E_{F1} - E_{V1} = -k_B T \ln (N_{A1} / N_{V2})$$

A Pn Heterojunction Diode in Thermal Equilibrium:

- **Built-in potential V_{bi}:**

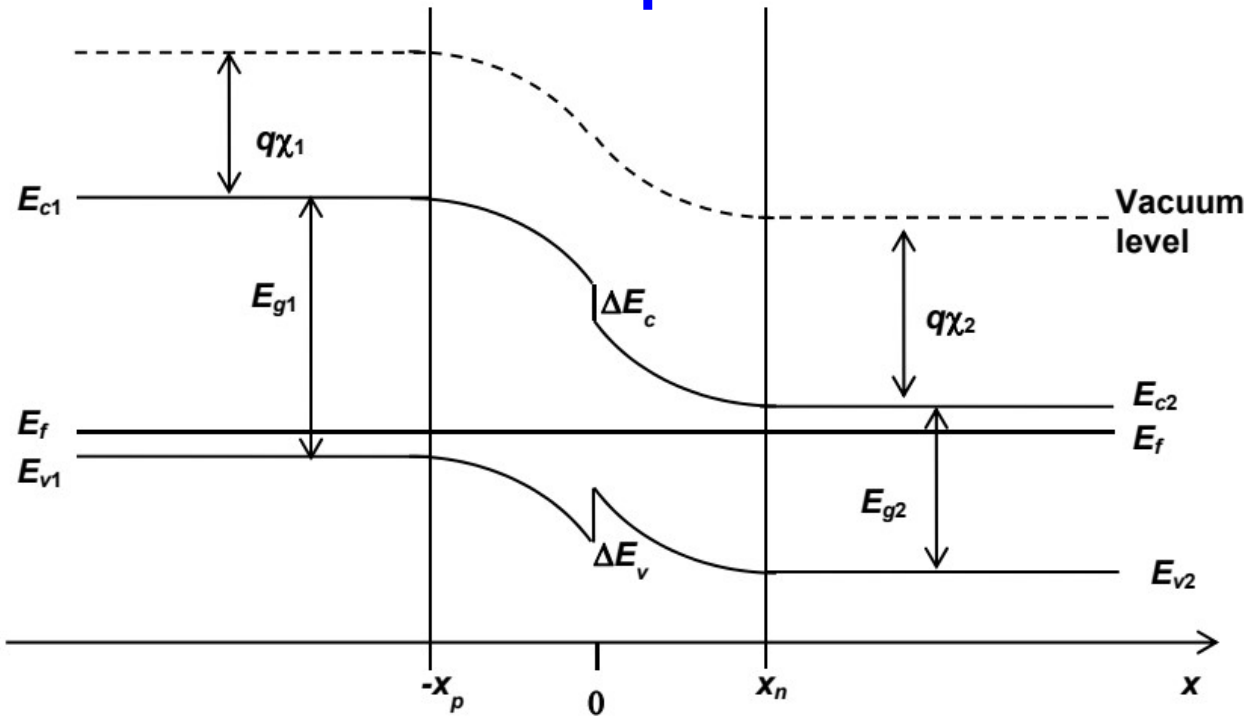
$$qV_{bi} = E_{f2} - E_{f1}$$

$$E_{v1} - E_{f1} = KT \ln\left(\frac{N_a}{N_{v1}}\right)$$

$$E_{f2} - E_{c2} = KT \ln\left(\frac{N_d}{N_{c2}}\right)$$

$$qV_{bi} = E_{f2} - E_{f1}$$

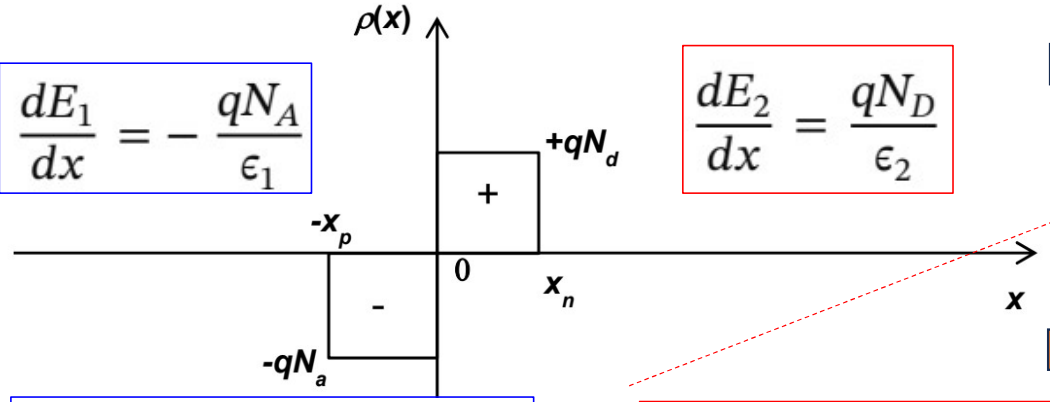
$$= E_{c2} - E_{v1} + KT \ln\left(\frac{N_a N_d}{N_{c2} N_{v1}}\right)$$



$$qV_{bi} = E_{g1} - \Delta E_c + KT \ln\left(\frac{N_a N_d}{N_{c2} N_{v1}}\right)$$

$$= E_{g2} + \Delta E_v + KT \ln\left(\frac{N_a N_d}{N_{c2} N_{v1}}\right).$$

Depletion Approximation:



$$\frac{dE_1}{dx} = -\frac{qN_A}{\epsilon_1}$$

$$\frac{dE_2}{dx} = \frac{qN_D}{\epsilon_2}$$

$$qN_d x_n = qN_a x_p \quad \text{유전상수도 다름}$$

$$D_1(0) = D_2(0) \Longrightarrow \epsilon_1 E_1(0) = \epsilon_2 E_2(0)$$

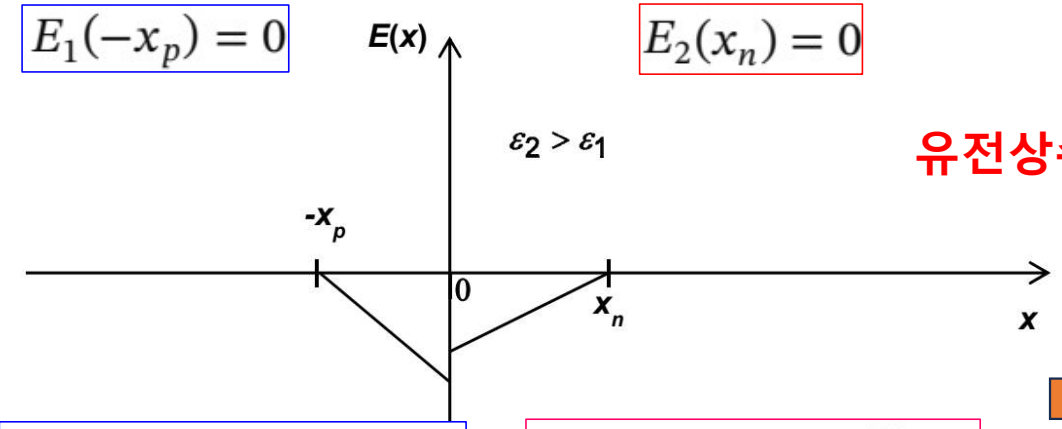
$$\rho(x) = \begin{cases} +qN_d & 0 < x \le x_n \\ -qN_a & -x_p \le x < 0 \\ 0 & \text{elsewhere} \end{cases}$$

$$\mathbf{E_1(x) = -\frac{qN_A}{\epsilon_1}(x + x_p)}$$

$$\mathbf{E_2(x) = -\frac{qN_D}{\epsilon_2}(x_n - x)}$$

$$E_1(-x_p) = 0$$

$$E_2(x_n) = 0$$



유전상수도 다름

$$E(x) = \begin{cases} +\dfrac{qN_d(x_n - x)}{\varepsilon_2} & 0 < x \le x_n \\ -\dfrac{qN_a(x + x_p)}{\varepsilon_1} & -x_p \le x < 0 \\ 0 & \text{elsewhere} \end{cases}$$

$$E_{max,1} = -\frac{qN_A x_p}{\epsilon_1}$$

$$E_{max,2} = -\frac{qN_D x_n}{\epsilon_2}$$

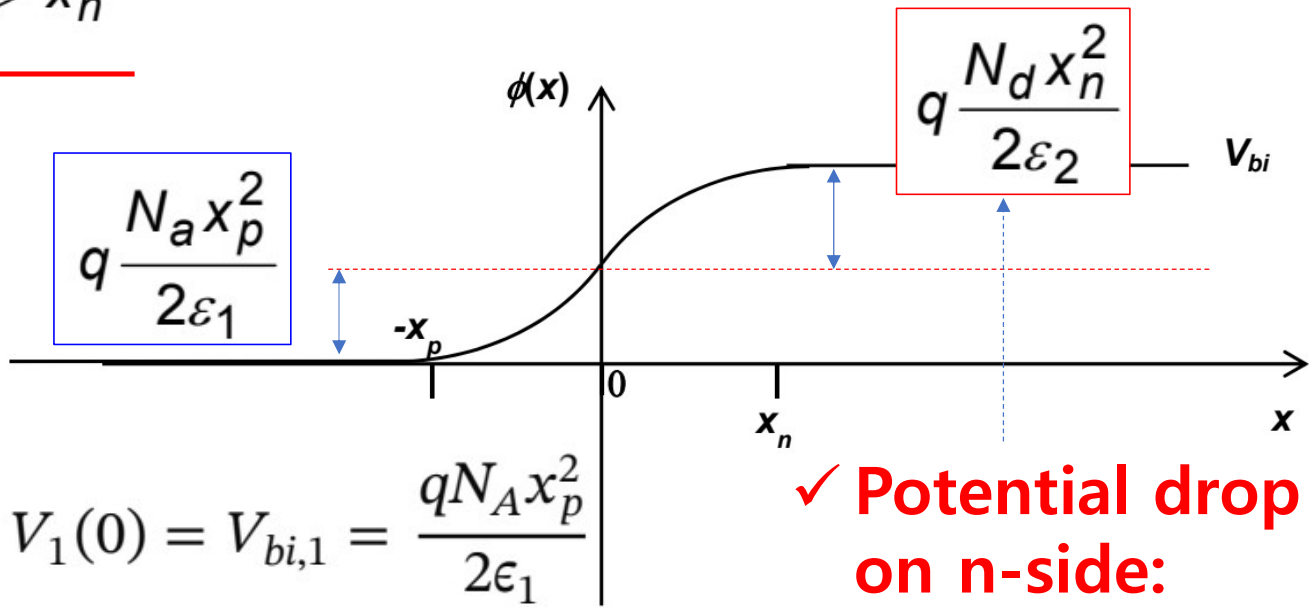
Depletion Approximation: Potential

$$\phi(x)=\begin{cases}\dfrac{qN_a x_p^2}{2\varepsilon_1}+\dfrac{qN_d\left(x_n x-x^2/2\right)}{\varepsilon_2} & 0<x\le x_n \\ \dfrac{qN_a\left(x+x_p\right)^2}{2\varepsilon_1} & -x_p\le x<0 \\ \dfrac{qN_a x_p^2}{2\varepsilon_1}+\dfrac{qN_d x_n^2}{2\varepsilon_2} & x>x_n \\ 0 & x<-x_p\end{cases}$$

$$V_2(x)=V_1(0)-\int_0^x E_2(x)\,dx$$

$$V_2(0)=V_1(0)=V_{bi,1}$$

$$V_1(x)=-\int_{-x_p}^{x} E_1(x)\,dx \qquad V_1(-x_p)=0$$



✓ **Potential drop on P-side:**

$$V_1(0)=V_{bi,1}=\frac{qN_A x_p^2}{2\epsilon_1}$$

➢ **Built-in potential V_{bi}:**

$$V_{bi}=q\frac{N_d x_n^2}{2\varepsilon_2}+q\frac{N_a x_p^2}{2\varepsilon_1}$$

✓ **Potential drop on n-side:**

Depletion Approximation:

$$V_{bi} = q\frac{N_d x_n^2}{2\varepsilon_2} + q\frac{N_a x_p^2}{2\varepsilon_1} \quad \Leftarrow \quad qN_d x_n = qN_a x_p \text{ (Charge Neutrality)}$$

$$x_p = \frac{N_D}{N_A} x_n$$

$$x_n = \frac{N_A}{N_D} x_p$$

- **Thicknesses of the depletion regions**

$$x_p = \left[\frac{2}{q}\varepsilon_1\varepsilon_2 \frac{N_d}{N_a}\frac{V_{bi}}{(\varepsilon_1 N_a + \varepsilon_2 N_d)}\right]^{\frac{1}{2}} \qquad x_n = \left[\frac{2}{q}\varepsilon_1\varepsilon_2 \frac{N_a}{N_d}\frac{V_{bi}}{(\varepsilon_1 N_a + \varepsilon_2 N_d)}\right]^{\frac{1}{2}}$$

- **Total thickness of depletion region**

$$W = x_n + x_p = \left[\frac{2}{q}\frac{\varepsilon_1\varepsilon_2}{(\varepsilon_1 N_a + \varepsilon_2 N_d)}\frac{(N_a + N_d)^2}{N_a N_d} V_{bi}\right]^{\frac{1}{2}}$$

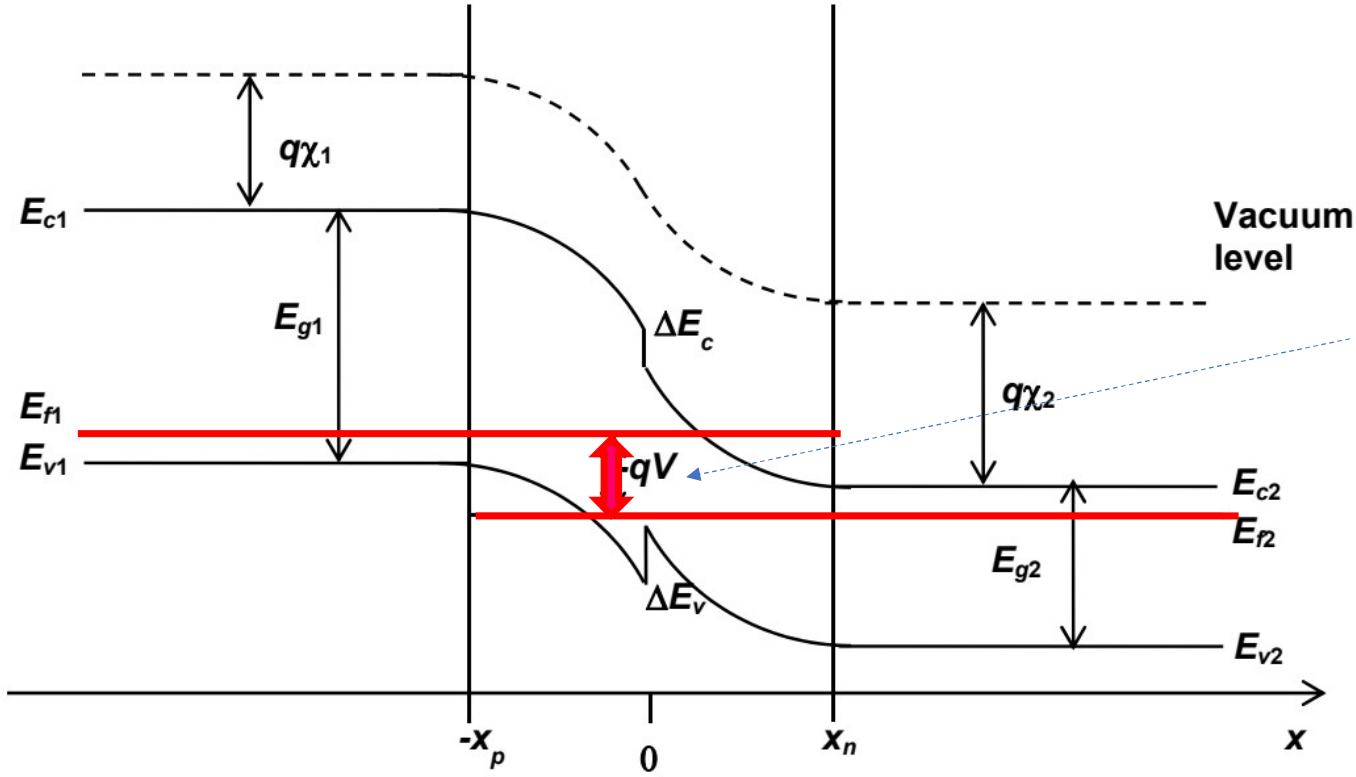
- **Charge (per unit area) on one side of the junction**

$$Q = \left[2q\frac{\varepsilon_1\varepsilon_2 N_a N_d}{(\varepsilon_1 N_a + \varepsilon_2 N_d)} V_{bi}\right]^{\frac{1}{2}}$$

$$qN_d x_n = qN_a x_p = Q$$

A Reverse Biased pn Heterojunction

Assume: V < 0 (reverse bias).



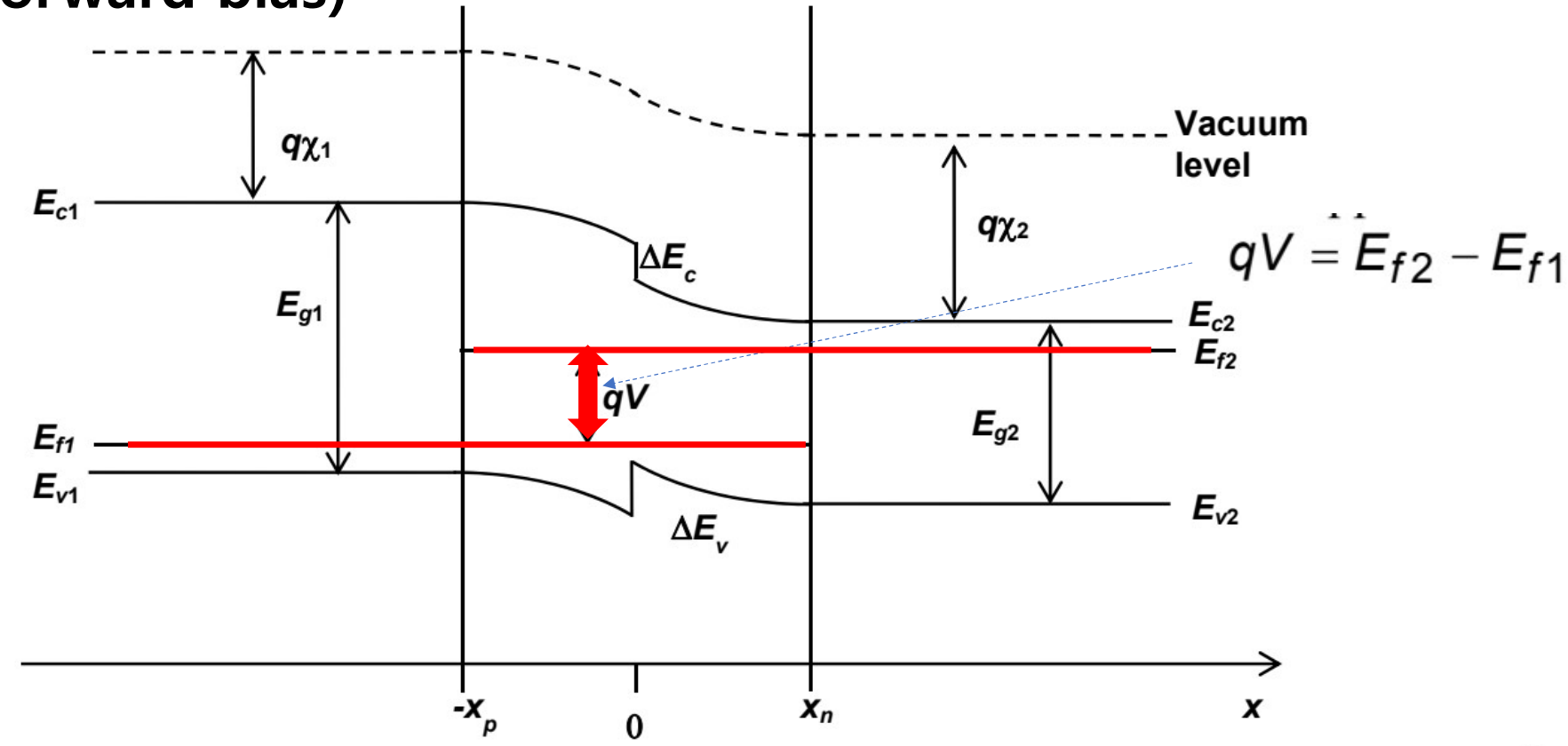
$$qV = E_{f2} - E_{f1}$$

$$x_p(V) = \left[\frac{2}{q}\varepsilon_1\varepsilon_2\frac{N_a}{N_d}\frac{(V_{bi}-V)}{(\varepsilon_1 N_a + \varepsilon_2 N_d)}\right]^{\frac{1}{2}}$$

$$x_n(V) = \left[\frac{2}{q}\varepsilon_1\varepsilon_2\frac{N_d}{N_a}\frac{(V_{bi}-V)}{(\varepsilon_1 N_a + \varepsilon_2 N_d)}\right]^{\frac{1}{2}}$$

A Forward Biased pn Heterojunction

V > 0 (Forward bias)



$$qV = E_{f2} - E_{f1}$$

$$x_p(V) = \left[\frac{2}{q}\varepsilon_1\varepsilon_2\frac{N_a}{N_d}\frac{(V_{bi}-V)}{(\varepsilon_1 N_a + \varepsilon_2 N_d)}\right]^{\frac{1}{2}}$$

$$x_n(V) = \left[\frac{2}{q}\varepsilon_1\varepsilon_2\frac{N_d}{N_a}\frac{(V_{bi}-V)}{(\varepsilon_1 N_a + \varepsilon_2 N_d)}\right]^{\frac{1}{2}}$$

Minority Carrier Injection and Current Flow

- Calculating current flow in Pn heterojunction diodes is complicated

➢ **Drift-Diffusion mode**

valid only if the band offsets ΔE_c and ΔE_v Ev are small (compared to KT).

➢ Essential assumption in the drift-diffusion:

- Carrier drift and diffusion in the n-doped and p-doped regions (not including the depletion regions): The main bottleneck for electron transport
- Transport through the depletion regions or at the heterointerface:

Not the main bottlenecks.

➔The main bottleneck to electron and hole transport in the device comes from the **quasi-neutral regions**

➔Through the depletion region (due to drift or diffusion): not the bottleneck.

➢ Assume that the electrons in the depletion region are in equilibrium with the electrons in the n-doped side (and share the same Fermi level) and holes in the depletion region are in equilibrium with the holes on the p-doped side (and share the same Fermi level). ➔Shockley boundary condition

➔Quasi-Equilibrium assumption in the depletion region

Minority Carrier Injection and Current Flow

➔ Quasi-Equilibrium assumption in the depletion region

$$n(-x_p) = N_{c1}\, e^{\left(E_{f2}(-x_p)-E_c(-x_p)\right)/KT}$$

$$= N_{c1}\, e^{\left(E_{f1}(-x_p)-E_c(-x_p)\right)/KT}\; e^{\left(E_{f2}(-x_p)-E_{f1}(-x_p)\right)/KT}$$

$$= \frac{n_{i1}^2}{N_a}\, e^{qV/KT} = n_{po}\, e^{qV/KT}$$

$$p(-x_p) = N_{v1}\, e^{\left(E_v(-x_p)-E_{f1}(-x_p)\right)/KT} = N_a$$

$$n(x_n) = N_{c2}\, e^{\left(E_{f2}(x_n)-E_c(x_n)\right)/KT} = N_d$$

$$p(x_n) = N_{v2}\, e^{\left(E_v(x_n)-E_{f1}(x_n)\right)/KT}$$

$$= N_{v2}\, e^{\left(E_v(x_n)-E_{f2}(x_n)\right)/KT}\; e^{\left(E_{f2}(x_n)-E_{f1}(x_n)\right)/KT}$$

$$= \frac{n_{i2}^2}{N_d}\, e^{qV/KT} = p_{no}\, e^{qV/KT}$$

$$n_{p0} = \frac{n_{i1}^2}{N_A}$$

$$p_{n0} = \frac{n_{i2}^2}{N_D}$$

$$pn = n_i^2 \exp\left(\frac{qV}{kT}\right)$$

$$\Delta n_p(-x_p) = n_p(-x_p) - n_{p0} = \frac{n_{i1}^2}{N_A}\left[\exp\left(\frac{qV}{kT}\right) - 1\right]$$

$$\Delta p_n(x_n) = p_n(x_n) - p_{n0} = \frac{n_{i2}^2}{N_D}\left[\exp\left(\frac{qV}{kT}\right) - 1\right]$$

Minority Carrier Injection and Current Flow

- The **generation-recombination rate** of these "**injected" electrons** (*minority carriers*) on the **P-side**

$$R_e(x) - G_e(x) = \frac{n(x) - n_{po}}{\tau_{e1}}$$

- **Assumption: no potential drop** across the **quasi-neutral p-doped region**
  - ➔The **electric field in this region** is **(almost) zero**
  - ➔The **electron current is entirely due to diffusion**, $J_e(x) = q D_{e1} \frac{\partial n(x)}{\partial x}$.
- **Continuity Equation**

$$\frac{\partial n}{\partial t} - \frac{1}{q}\frac{\partial}{\partial x} J_e(x) = G_e(x) - R_e(x)$$

$$\Rightarrow \frac{\partial n}{\partial t} = 0 \text{ (no time dependence in steady state)}$$

$$\Rightarrow D_{e1} \frac{\partial^2 n(x)}{\partial x^2} = \frac{n(x) - n_{po}}{\tau_{e1}} \quad \Rightarrow \quad \frac{\partial^2 n(x)}{\partial x^2} = \frac{n(x) - n_{po}}{L_{e1}^2}$$

$$L_{e1} = \sqrt{D_{e1}\,\tau_{e1}} = \text{Minority carrier diffusion lenght}$$

Minority Carrier Injection and Current Flow

- **For the holes injected into the n-side the corresponding equation**

$$D_n \frac{d^2\Delta n_p(x)}{dx^2} - \frac{\Delta n_p(x)}{\tau_n} = 0 \Rightarrow \frac{\partial^2 p(x)}{\partial x^2} = \frac{p(x) - p_{no}}{L_{h2}^2} \qquad L_{h2} = \sqrt{D_{h2}\,\tau_{h2}}$$

- ✓ **Consider first the electrons on the P-side:** $n(-x_p) = n_{po}\, e^{qV/KT}$
- ✓ **Assume: At the left most metal contact** the **recombination time is extremely fast** : The **electron density** must **equal the equilibrium density** $n(-W_p) = n_{po}$

- **Solution for the electron density**

$$\Rightarrow \frac{\partial^2 n(x)}{\partial x^2} = \frac{n(x) - n_{po}}{L_{e1}^2} \qquad -W_p \le x \le -x_p$$

$$n(x) = n_{po} + n_{po} \frac{\sinh\left(\frac{W_p + x}{L_{e1}}\right)}{\sinh\left(\frac{W_p - x_p}{L_{e1}}\right)} \left(e^{\frac{qV}{KT}} - 1\right)$$

$$\frac{d^2\Delta n}{dx^2} - \frac{\Delta n}{L^2} = 0$$

$$\Delta n(x) = A\sinh\left(\frac{x}{L}\right) + B\cosh\left(\frac{x}{L}\right)$$

$$n_p(x') = \frac{n_{i1}^2}{N_A} + \frac{n_{i1}^2}{N_A}\left[\exp\left(\frac{qV}{kT}\right) - 1\right]\exp\left(-\frac{x'}{L_n}\right)$$

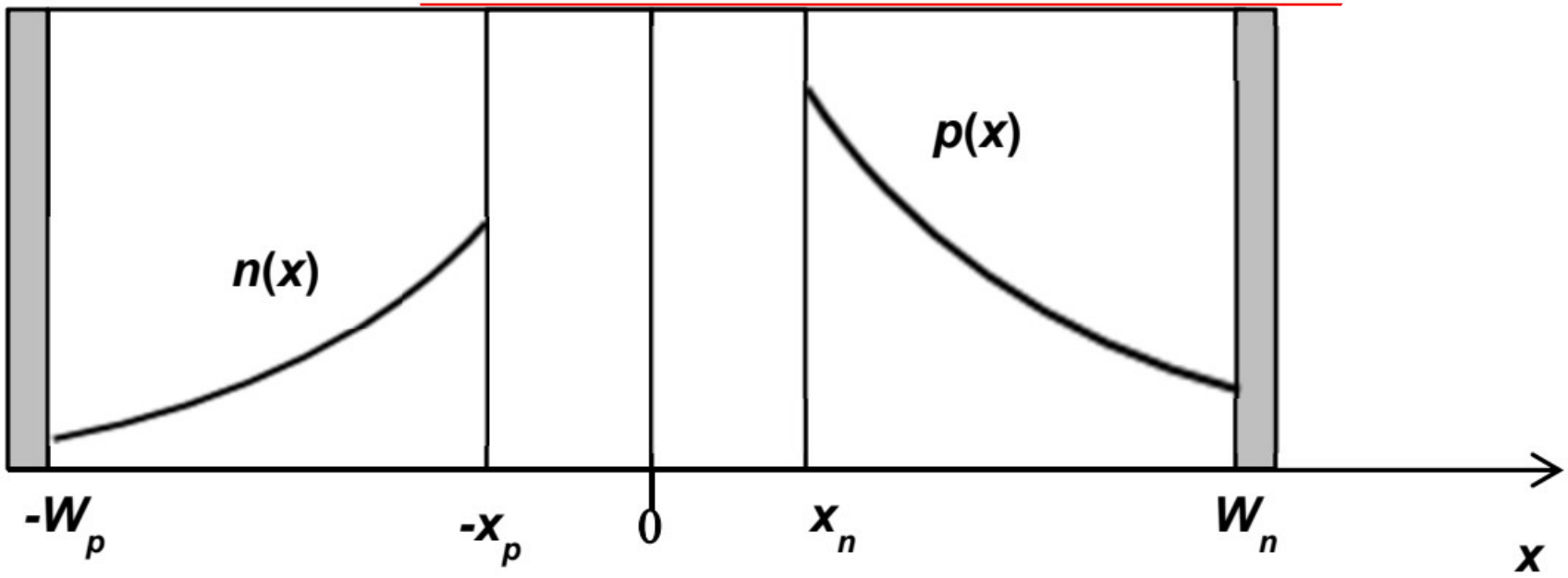
Minority Carrier Injection and Current Flow

- **For the holes on the n-side**

$$D_p \frac{d^2 \Delta p_n(x)}{dx^2} - \frac{\Delta p_n(x)}{\tau_p} = 0$$

$$p(x) = p_{no} + p_{no} \frac{\sinh\left(\frac{W_n - x}{L_{h2}}\right)}{\sinh\left(\frac{W_n - x_n}{L_{h2}}\right)} \left(e^{\frac{qV}{KT}} - 1 \right) \qquad x_n \le x \le W_n$$

$$p_n(x'') = \frac{n_{i2}^2}{N_D} + \frac{n_{i2}^2}{N_D} \left[\exp\left(\frac{qV}{kT}\right) - 1 \right] \exp\left(-\frac{x''}{L_p}\right)$$



Minority Carrier Injection and Current Flow

- Electron current density J_e (due to diffusion) on the p-side

$$J_e(x) = qD_{e1}\frac{\partial n(x)}{\partial x} = q\frac{n_{i1}^2}{N_a}\frac{D_{e1}}{L_{e1}}\frac{\cosh\left(\frac{W_p + x}{L_{e1}}\right)}{\sinh\left(\frac{W_p - x_p}{L_{e1}}\right)}\left(e^{\frac{qv}{KT}} - 1\right) \qquad -W_p \le x \le -x_p$$

- Hole current density on the n-side

$$J_h(x) = -qD_{nh}\frac{\partial p(x)}{\partial x} = q\frac{n_{i2}^2}{N_d}\frac{D_{h2}}{L_{h2}}\frac{\cosh\left(\frac{W_n - x}{L_{n2}}\right)}{\sinh\left(\frac{W_n - x_n}{L_{n2}}\right)}\left(e^{\frac{qv}{KT}} - 1\right) \qquad x_n \le x \le W_n$$

- Total current density
  = the sum of the electron and the hole density

$$J_T = J_e(x) + J_h(x)$$

Minority Carrier Injection and Current Flow

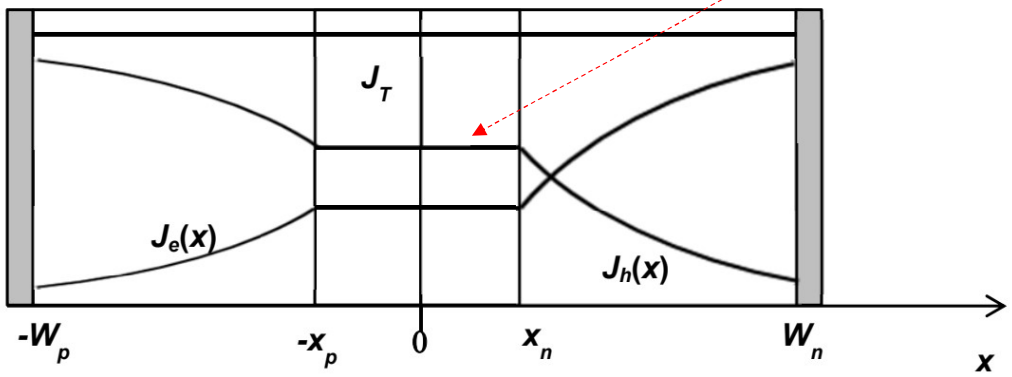
- **Assume:**
  - ✓ Ignores recombination and generation processes inside the depletion region
  - ✓ The electron and hole current density must be constant throughout the depletion region,

$$J = J_n(x' = 0) + J_p(x'' = 0)$$

$$J_n(x' = 0) = qD_n \left.\frac{dn_p(x')}{dx'}\right|_{x'=0}$$

$$J_p(x'' = 0) = -qD_p \left.\frac{dp_n(x'')}{dx''}\right|_{x''=0}$$

$$\begin{cases} J_e(-x_p) = J_e(x_n) = J_e(x) & \{-x_p \le x \le x_n\} \\ J_h(-x_p) = J_h(x_n) = J_h(x) & \{-x_p \le x \le x_n\} \end{cases}$$



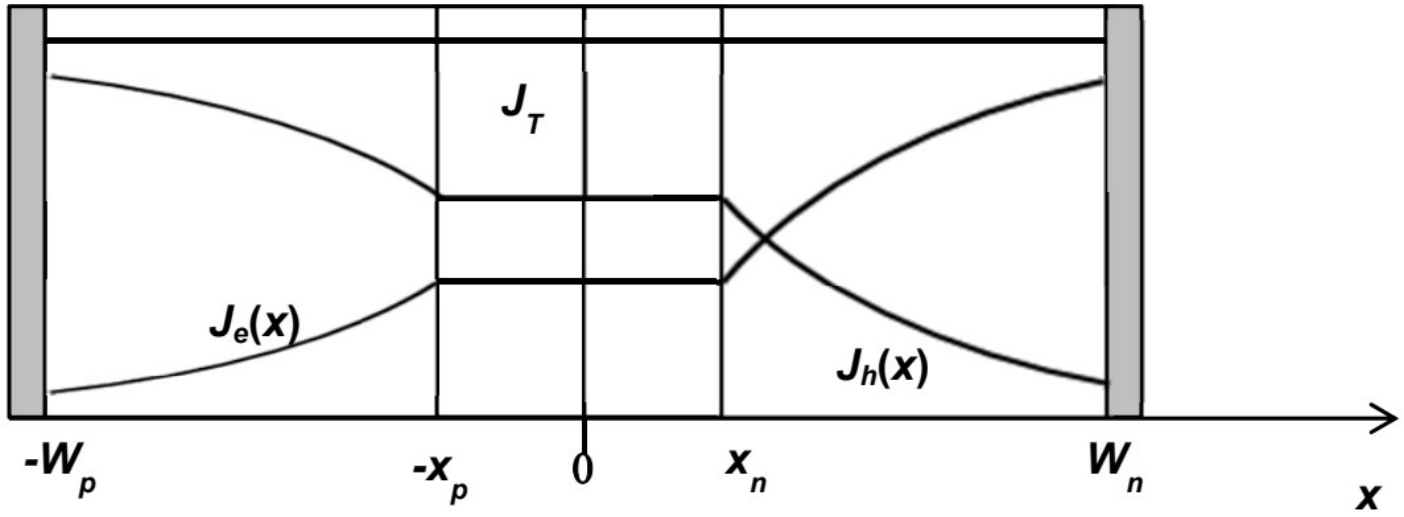
$$J_e(-x_p) = q\frac{n_{i1}^2}{N_a}\frac{D_{e1}}{L_{e1}}\coth\left(\frac{W_p - x_p}{L_{e1}}\right)\left(e^{\frac{qv}{KT}} - 1\right)$$

$$J_h(x_n) = q\frac{n_{i2}^2}{N_d}\frac{D_{h2}}{L_{h2}}\coth\left(\frac{W_n - x_n}{L_{h2}}\right)\left(e^{\frac{qv}{KT}} - 1\right)$$

Minority Carrier Injection and Current Flow

$$J_T = q\left\{\frac{n_{i1}^2}{N_a}\frac{D_{e1}}{L_{e1}}\coth\left(\frac{W_p - x_p}{L_{e1}}\right) + \frac{n_{i2}^2}{N_d}\frac{D_{h2}}{L_{h2}}\coth\left(\frac{W_n - x_n}{L_{h2}}\right)\right\}\left(e^{\frac{qv}{KT}} - 1\right)$$

$$I = AJ_T = I_o\left(e^{\frac{qv}{KT}} - 1\right)$$



$$I_o = qA\left\{\frac{n_{i1}^2}{N_a}\frac{D_{e1}}{L_{e1}}\coth\left(\frac{W_p - x_p}{L_{e1}}\right) + \frac{n_{i2}^2}{N_d}\frac{D_{h2}}{L_{h2}}\coth\left(\frac{W_n - x_n}{L_{h2}}\right)\right\}$$

Majority Carrier Dynamics and Quasi-Neutrality:

$$\Delta p(x) = p(x) - N_a = \Delta n(x) = n(x) - n_{po} \quad -W_p \le x \le -x_p$$

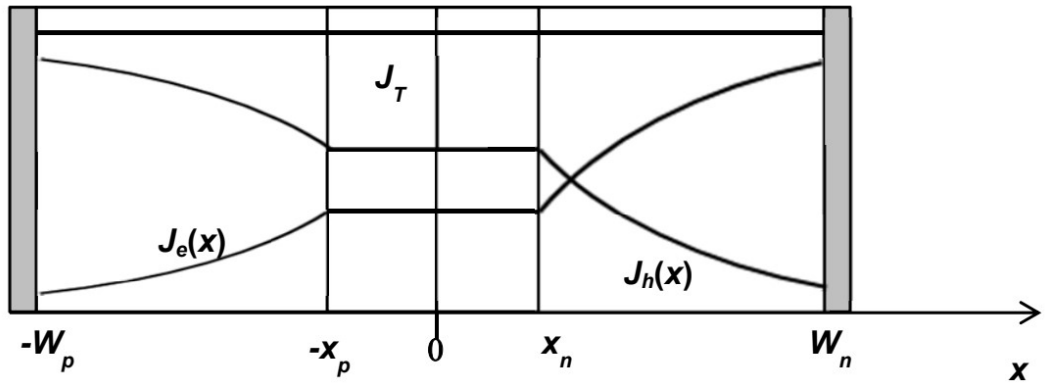
$$\Delta n(x) = n(x) - N_d = \Delta p(x) = p(x) - p_{no} \quad x_n \le x \le W_n$$

- The **excess majority carrier concentration**
  = The excess minority carrier concentration
  to maintain charge neutrality
- **On the p-side the hole currents**

$$J_h(x) = qp(x)\mu_{h1}E(x) - qD_{h1}\frac{\partial p(x)}{\partial x}$$
$$= J_T - J_e(x)$$

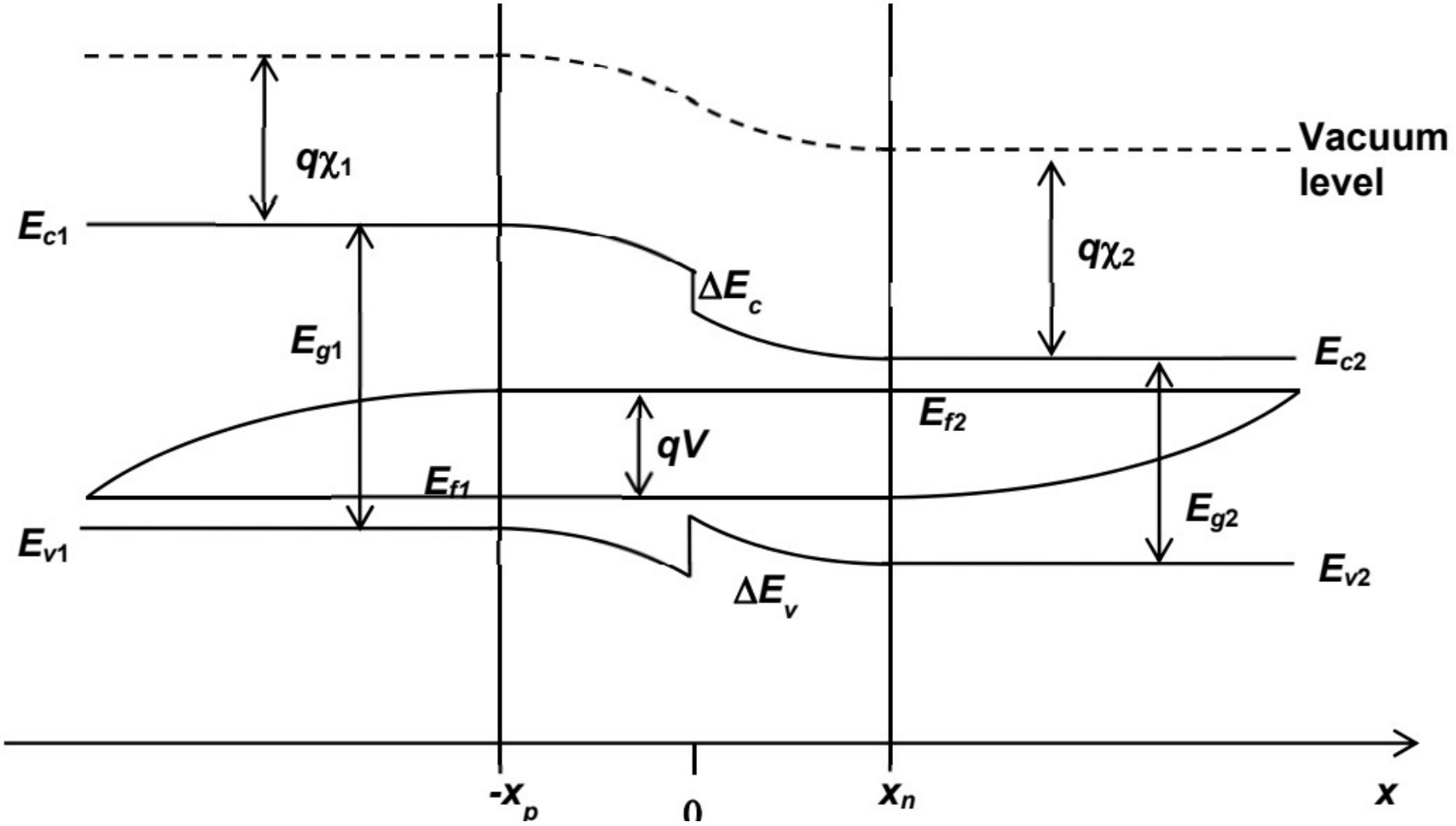
- **On the n-side, the electron curren**

$$J_e(x) = qn(x)\mu_{e2}E(x) + qD_{e2}\frac{\partial n(x)}{\partial x}$$
$$= J_T - J_h(x)$$

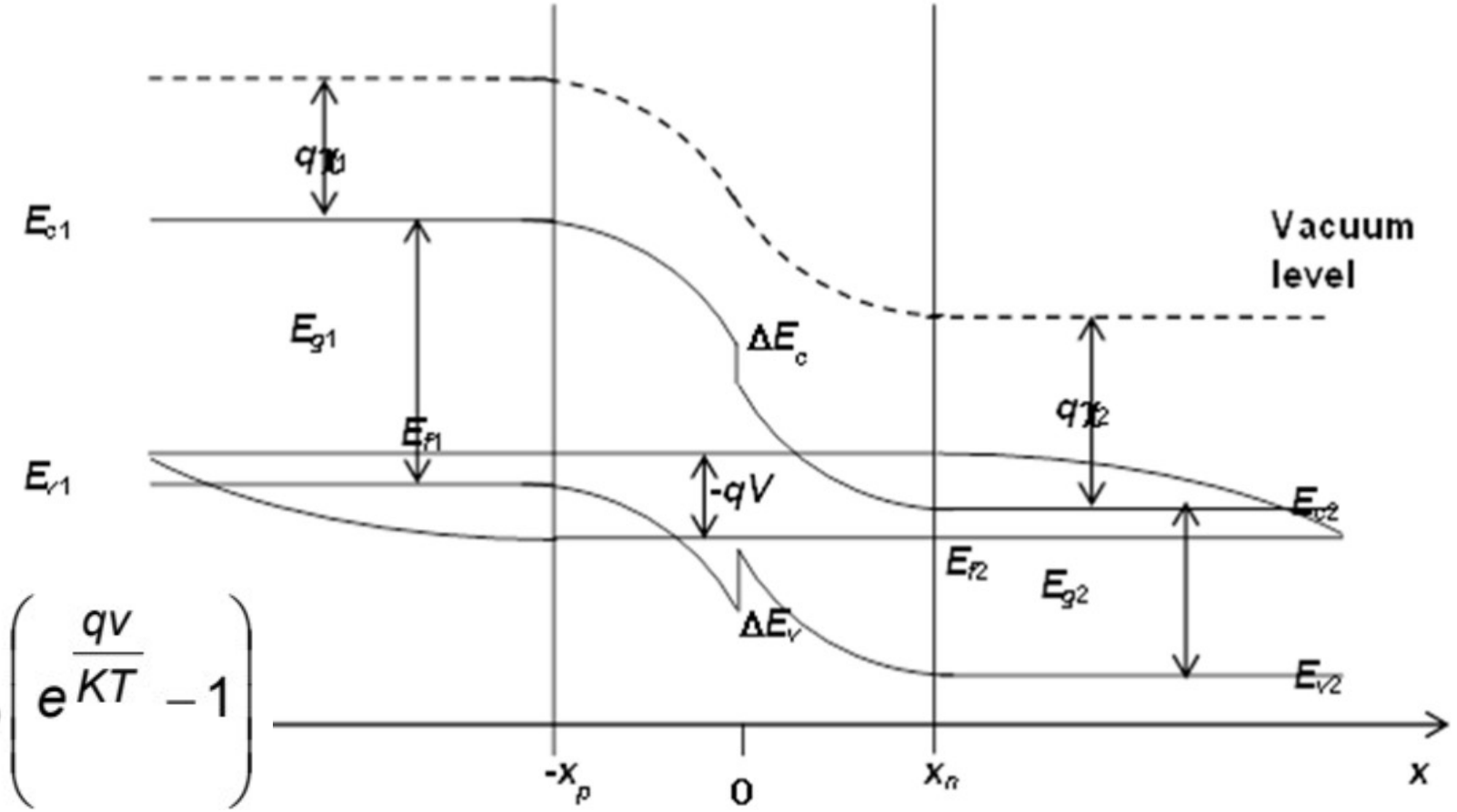


Electron and hole Fermi levels

through the entire device in forward bias



➢ Current Flow in Reverse Biased Pn Heterojunction:



$$I = I_o\left(e^{\frac{qv}{KT}} - 1\right)$$

$$I_o = qA\left\{\frac{n_{i1}^2}{N_a}\frac{D_{e1}}{L_{e1}}\coth\left(\frac{W_p - x_p}{L_{e1}}\right) + \frac{n_{i2}^2}{N_d}\frac{D_{h2}}{L_{h2}}\coth\left(\frac{W_n - x_n}{L_{h2}}\right)\right\}$$

Generation and Recombination in the Depletion Region

- **Generation and recombination in the depletion region** also **contributes to the total diode current in both forward and reverse bias.**
- **Electron current inside the depletion region.**
  - **Continuity equation in steady state**

$$-\frac{1}{q}\frac{\partial}{\partial x}J_e(x) = G_e(x) - R_e(x)$$

  - **Integrating over the depletion region**

$$J_e(x_n) - J_e(-x_p) = q\int_{-x_p}^{x_n}[R_e(x) - G_e(x)]dx$$

- **For the hole current,**

$$J_h(-x_p) - J_h(x_n) = q\int_{-x_p}^{x_n}[R_h(x) - G_h(x)]dx$$

◆**For the total diode current,**

add the electron and hole currents **at one edge of the depletion region**

$$J_T = q\left\{\frac{n_{i1}^2}{N_a}\frac{D_{e1}}{L_{e1}}\coth\left(\frac{W_p - x_p}{L_{e1}}\right) + \frac{n_{i2}^2}{N_d}\frac{D_{h2}}{L_{h2}}\coth\left(\frac{W_n - x_n}{L_{h2}}\right)\right\}\left(e^{\frac{qv}{KT}} - 1\right) + q\int_{-x_p}^{x_n}[R_e(x) - G_e(x)]dx$$

✓ Current in the external circuit

$$I = AJ_T = I_o\left(e^{\frac{qv}{KT}} - 1\right) + qA\int_{-x_p}^{x_n}[R_e(x) - G_e(x)]dx$$

- **In the case** of **trap-assisted recombination-generation** (the dominant mechanism in silicon pn diodes),

$$G_e(x) - R_e(x) \propto \left(np - n_i^2\right)$$

- **Inside the depletion region,**

$$np = n_i^2 e^{(E_{fc} - E_{fh})/KT} \qquad G_e(x) - R_e(x) \propto \left(e^{\frac{qv}{KT}} - 1\right)$$